
FORKSERVE: PREFILL PRUNING AND DISAGGREGATED KV FOR AGENTIC LLM SERVING

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ABSTRACT

Agentic LLM serving expands one trunk into several short residuals and then keeps a single committed path. The engines in wide use still charge the paths that the harness will drop. Automatic prefix caching and radix prefix sharing reuse a trunk only after the tokens exist and a hash hits, so the first fan-out copies that trunk once per child. Prefill–decode disaggregation separates time-to-first-token from decode latency, yet the connector transfers every page the prefiller computed, including residuals that are later aborted. Decode-time pruners—early-stopping self-consistency (ESC), Speculative Rejection, and dynamic parallel tree search (DPTS)—score a child only after that child has been prefilled and a prefix has been generated.

ForkServe prunes that fan-out before a disaggregated connector moves it. Each residual receives one action, in increasing cost. A block-hash probe reuses published pages. A repeated loop is draft-skipped before a kernel starts. Any other low-score residual is prefilled and decoded for a short probe, and continues to the budget only if the generated prefix clears the threshold. The connector inserts only the miss tail of a residual that is kept; aborted pages stay on the prefiller. A copy-on-write context tree is the substrate: `fork` aliases the parent’s KV pages, and abort releases the residual alone, so decode attends the committed spine. Longest-common-prefix commit binds the prompt presented by the harness. Speculative tokens never enter the sampler, so committed outputs match a baseline that prefills only committed tokens. A scheduler admits speculative prefill only from the budget left after committed requests meet their latency margins.

On Qwen3-8B GSM8K ($n=16$, $D=512$), $\tau_{\text{pre}}=0.15$ in front of ESC, Speculative Rejection, or DPTS admits 12 of 16 residuals, cuts computed prefill from 5008 to 3848 tokens (-23%), keeps 16/16 any-finished at $k=16$, and reduces ESC time from 17.05 s to 14.14 s. That kept span is what disaggregation transfers. $\tau_{\text{pre}}=0.45$ reaches 10.93 s at the same 16/16 and drops two questions at $k=4$ and $k=8$. On Qwen3-8B MATH-500, dropping every residual below τ cuts any-finished accuracy from 20/32 to 13/32 at fan-out 16; a 128-token probe that continues only when the generated prefix clears the threshold restores 20/32 and cuts end-to-end time from 104.1 s to 72.2 s. On the Qwen3-14B GSM8K forest the committed spine matches prefix-cache accuracy at $0.72\times$ the median peak KV. The earlier blanket-skip cost model, which drops every low-score residual before prefill, cuts modeled fan-out prefill by 74.72% and raises pool concurrency by 35.53%; the GPU rule above replaces that skip.

1 INTRODUCTION

LLM serving began as a single-turn problem: a prompt arrives, the engine prefills, and tokens are streamed [47, 17, 53]. Production agents—coding assistants [44, 39], computer-use systems [55], and multi-agent workflows [42, 14]—execute programs of LLM calls interleaved with tools [46, 50]. Tree-of-Thoughts and MCTS expand sibling thoughts that share a prefix and diverge in a short residual [45, 26]. The serving unit is a *branching execution*: one trunk of length L , k residuals ℓ_i , and one committed spine.

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Three families of systems already sit on that execution, and each charges the branch at a fixed stage (§2). Automatic prefix caching (APC) [17] and RadixAttention [53] reuse a trunk after a full block has been published, so a same-batch fan-out still materializes k copies of L . Prefill–decode disaggregation [54, 32] isolates TTFT from inter-token latency and then transfers every page the prefiller computed, aborted residuals included. Decode-time pruning scores a child from tokens that are already resident: ESC [21] waits for an agreeing window, Speculative Rejection [31] waits for a partial-reward horizon, and DPTS [19] waits for a mini-step. In each case the residual has already been prefilled. Program-, session-, and workflow-level schedulers [25, 16, 20, 43, 36, 23, 48] raise the atom above the request and still issue the next call as a unique continuation.

Once KV retention has removed trunk recompute [20, 43], the remaining TTFT term is residual-suffix prefill of children the harness can name before the observation returns.

This work. We present ForkServe (Figure 1). Prefill pruning decides which residual is computed; disaggregation then moves only that residual. Three mechanisms implement the rule.

(1) A copy-on-write context tree (§3.2). Each session is a tree of committed and speculative nodes. `fork` creates a child that inherits the parent’s pages by reference; writes allocate residual blocks. `abort` drops a subtree and decrements residual refcounts only, so live occupancy collapses to the spine $M_{\text{spine}} = b(L + \ell_*)$. APC discovers the same sharing only after the tokens exist; the alias exists at `fork`, which is why a pruned residual never allocates a second trunk for the connector to ship.

(2) Prefill pruning and a gated connector (§3.4). APP assigns each residual ρ_i one action $a_i \in \mathcal{A}$ and a pair (W_i, X_i) . A published prefix of length m_i is retrieved: $m_i = \ell_i$ is a hash skip and $0 < m_i < \ell_i$ prefills the miss tail. A miss has no row in the index. A repeated loop is draft-skipped before a kernel starts. Any other low-score residual is prefilled and decoded for a probe of t_0 tokens, and continues to the budget only if the generated prefix clears the threshold. Index 0 is forced into $\mathcal{K}$. Baseline disaggregation inserts the span it computed, including the k -way trunk [54, 32]. APP inserts only the miss tail of $\mathcal{K}$. The same threshold sits in front of ESC, Speculative Rejection, or DPTS: those hosts still decide when to stop decoding, and the connector never receives a child the stage has already rejected. LCP commit binds the observed prompt; speculative tokens never enter the sampler.

(3) The spine on the critical path (§3.5). Decode slots are $C = \lfloor H/M_{\text{spine}} \rfloor$. On tick t the budget splits as $B_t = B_t^c + B_t^s$: B_t^c decodes the committed spine and B_t^s prefills the next winner’s miss tail in the same step. A known suffix with $p(w)=1$ and $T_{\text{pre}}(|w|) \leq T_{\text{idle}}(u)$ is pinned during the tool pause; the observation arrives as the unmatched tail.

§4 leads with the GPU pruning rule and then the earlier cost model it replaces. On Qwen3-8B GSM8K, $\tau_{\text{pre}}=0.15$ in front of ESC, Speculative Rejection, or DPTS admits 12 of 16 residuals, cuts computed prefill by 23% (5008 to 3848), keeps 16/16 any-finished at $k=8$ and $k=16$, and reduces ESC time from 17.05 s to 14.14 s; $\tau_{\text{pre}}=0.45$ is faster at $k=16$ (10.93 s, still 16/16) but drops two questions at smaller fan-out (Tables 8 and 9). On Qwen3-8B MATH-500, a blanket skip of every low score drops any-finished accuracy from 20/32 to 13/32 at $k=16$; a 128-token probe restores 20/32 and cuts end-to-end time from 104.1 s to 72.2 s (Table 7). On a three-turn Tree-of-Thoughts, Fork-

Serve+ prefills 48 of 192 residuals and expands $7.9\text{--}10.6\times$ faster than APC (Tables 11 and 12). The Qwen3-14B forest still matches APC accuracy at $0.72\times$ median peak KV. An earlier control-plane model that blanket-skips every low-score residual cuts fan-out prefill by 74.72% against APC and raises SLO-feasible concurrency by 35.5%; §4.2 replaces that skip. A six-model forest ($n=40$) keeps GSM8K peak KV at $0.72\text{--}0.75\times$ APC.

2 BACKGROUND AND MODEL

2.1 Prefill, Decode, and Idle

Autoregressive inference has two mechanically different phases [47, 54, 32]. *Prefill* consumes the prompt in parallel and is compute-bound; *decode* emits one token (or a small chunk) per step and is memory-bandwidth bound. PagedAttention [17] stores KV in non-contiguous pages; RadixAttention [53] indexes pages by token prefix so concurrent requests that share a system prompt hit a common trunk. Two consequences matter for agents. First, next-turn TTFT is the prefill of tokens that miss the cache, plus queuing. Second, a memory-bound decode batch rarely saturates the tensor cores that a concurrent chunked prefill could use [2]. Agent serving adds a third fact: the prompt of turn $i+1$ is a *delta* on turn i , applied after a tool pause of highly variable length [20, 43].

2.2 Traditional Methods on a Fan-out

We use *traditional methods* for the mechanisms a production stack already applies to one fan-out: prefix caching, hash-indexed prefill, prefill–decode disaggregation, and decode-time pruning. They share the trunk or they drop a child, and they do so at a stage that is fixed in advance. Table 1 is that stage.

First-expand cost. APC indexes a full block by $h = \text{hash}(\text{parent}, \text{tokens}, \text{extra})$ after those tokens exist [17]; RadixAttention walks the same prefix [53]. A same-batch expansion therefore misses and clones k trunks, $W_{\text{APC}} = kL + \sum_i \ell_i$, while a later replay of x_* is a hit ($W=0$). Baseline disaggregation [54, 32] transfers that span,

$$X_{\text{PD}} = kL + \sum_i \ell_i, \quad (1)$$

so $X_{\text{PD}} = W_{\text{APC}}$ on the first expand. ESC [21], Speculative Rejection [31], and DPTS [19] evaluate a child only after its prefill and a generated prefix, hence after W and, under disaggregation, after X .

2.3 Fan-out Model

Definition 1 (Branching step). A parent u holds a committed trunk x_u with $|x_u|=L$. A fan-out is a set of residuals $\{\rho_i\}_{i=1}^k$, $|\rho_i|=\ell_i$. The bound winner $\star$ leaves the spine

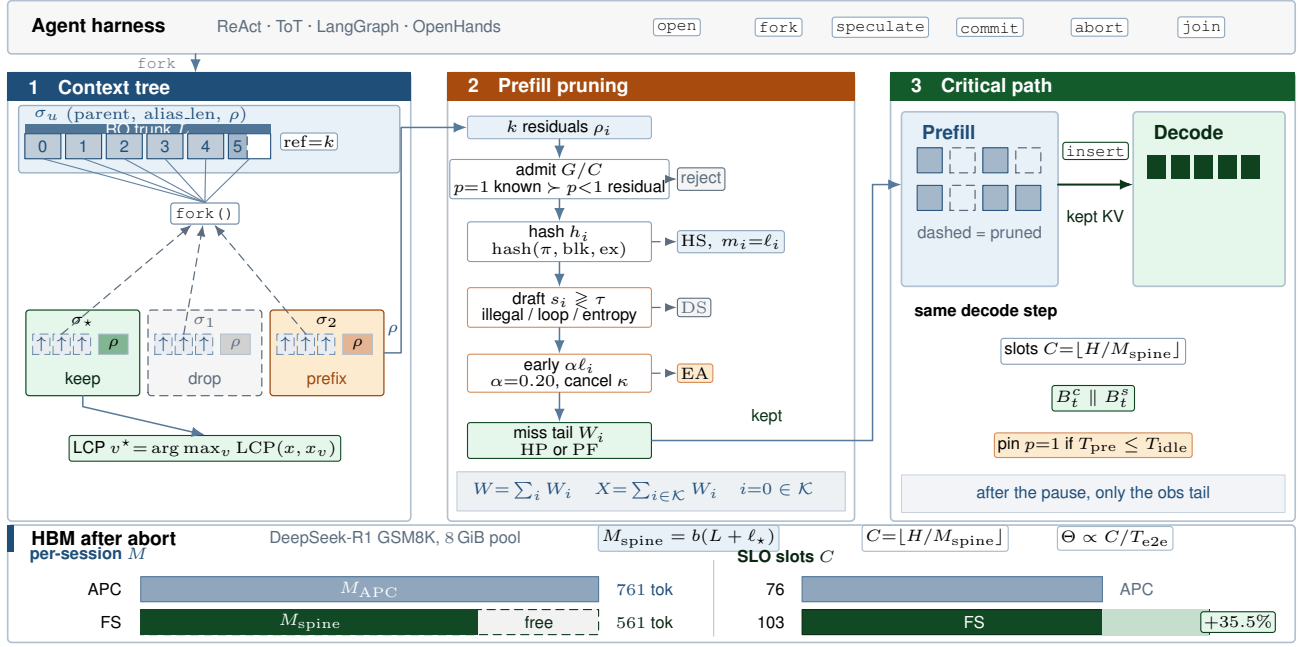


Figure 1. ForkServe architecture. (1) σ_u holds one read-only trunk; `fork` aliases it in $O(1)$; LCP binds the winner. (2) A residual exits at the first decisive action. $m_i = \ell_i$ is HS ($W=0$); $0 < m_i < \ell_i$ is HP; a repeated loop is DS; the first $\alpha \ell_i$ failing is EA; otherwise PF prefills ℓ_i on the alias, and a low score that is not a loop is probed for t_0 tokens before it may continue. Index 0 stays in $\mathcal{K}$, and X sums W_i only over that kept set. (3) Slots are $C = \lfloor H/M_{\text{spine}} \rfloor$; B_t^c runs beside B_t^s , and a known suffix is pinned when $p=1$ and $T_{\text{pre}} \leq T_{\text{idle}}$. After abort the live occupancy is the spine; the same pool admits 76→103 sessions.

Table 1. Where each traditional method pays for one fan-out of k residuals on a trunk of length L . “First expand” is the GPU prefill issued before any child has published a block. “Rejection” is the earliest point at which a non-winner can leave. ForkServe is the stage order of §3.4.

Method	First expand	Rejection	Connector
Recompute	k clones of L	end of the request	none
APC, radix, hash prefill	k clones of L	LRU after a block is published	none; a later replay of the winner is a hit
P/D disaggregation	k clones of L	after insert	$kL + \sum_i \ell_i$
ESC, Speculative Rejection, DPTS	every residual	after a generated prefix	KV of every prefilled child
ForkServe	one alias of L	loop before the kernel; other low scores after a t_0 probe	miss tail of $\mathcal{K}$

$$x_* = x_u \parallel \rho_*, |x_*| = L + \ell_*.$$

Let b be KV bytes per token and P the page size. The three occupancies of one step are

$$\begin{aligned} M_{\text{clone}} &= b(kL + \sum_i \ell_i), \\ M_{\text{CoW}} &= b(L + \sum_i \ell_i), \\ M_{\text{spine}} &= b(L + \ell_*). \end{aligned} \quad (2)$$

M_{clone} is the first-expand footprint of recompute, APC, and baseline disaggregation; M_{CoW} is one shared trunk with every residual live; M_{spine} is the footprint after abort. W is the token count issued to a prefill kernel before $*$ is known; X is the token count inserted on the P/D connector. Table 2 fixes the remaining symbols.

3 DESIGN

ForkServe is placed between the agent runtime and the LLM engine (Figure 1). The harness announces structure; the control plane admits, prunes, binds, and retains; the backend moves pages and tokens.

3.1 API

<code>open(session)</code>	Create a root; prefill system+user.
<code>fork(u, bid, tok?)</code>	Child of u ; optional known suffix.
<code>speculate(v, pri)</code>	Mark v eligible for speculative prefill.
<code>commit(u, tokens)</code>	Bind by LCP; start decode.
<code>join(children, pol)</code>	Merge fan-out (all/first/kofn).
<code>abort(v) / close</code>	Drop a subtree; end the session.

Table 2. Notation for Definition ?? and for (5)–(7).

Symbol	Domain	Meaning
L, ℓ_i, ℓ_*	$\mathbb{N}$	trunk; residual i ; committed residual
ρ_i, x_u	tokens	residual of child i ; committed trunk
h_i	digest	hash(parent, $\rho_i[0:P)$, extra)
m_i	tokens	full-page prefix of ρ_i in H , steps of P
s_i, s'_i, τ	$[0, 1]$	score of ρ_i ; score of the probe y_i ; threshold
t_0, D, d_i	tokens	probe length; decode budget; length assigned to i
α	$(0, 1]$	early-abort prefix fraction (0.20)
p_i, q_i	$[0, 1]$	admit probability; residual-prefix hit rate
$a_i \in \mathcal{A}$	$\mathcal{A}$	HS, HP, DS, EA, PF ((5))
Λ, F_i	set, $\{0, 1\}$	repeated-loop index set; early-abort predicate
$\mathcal{K}$	$\subseteq [k]$	residuals that remain after admission
W_i, X_i	tokens	GPU miss tail; connector payload
b, P	bytes, tok	KV bytes per token; page size
$B_t = B_t^c + B_t^s$	tokens	committed and speculative budget on tick t

`commit` is the only verb on the critical path of user-visible tokens.

3.2 Forkable Branch State

Definition 2 (Context tree). A session is a rooted tree $T = (V, E)$. Node u stores a token sequence x_u such that $x_{\pi(u)} \preceq x_u$, a residual $\delta_u = x_u[|x_{\pi(u)}|:]$, a mode $\rho_u \in \{\text{SPEC}, \text{COMMIT}, \text{IDLE}, \text{DEAD}\}$, a generation g_u , and a logical page table $\sigma_u = (\text{parent}, \text{alias_len}, \rho_u^{\text{pages}})$.

Invariants: (i) prefix; (ii) at most one committed child per node except during join; (iii) speculative nodes are invisible to sampling; (iv) a page’s refcount equals the number of non-dead nodes that map it. `fork`(u, bid, x^k) copies two words (parent $\leftarrow \sigma_u$, alias_len $\leftarrow |x_u|$) and increfs the aliased frames— $O(1)$ in L . A write that hits ref > 1 or a read-only tail CoW-copies only the valid rows. Figure 2 is that pool. A multi-level fan-out is a *sequence* of CoWs (Figure 4): each edge aliases the parent and allocates only +r; live paths share every page on the common prefix; abort is residual-only, and a subtree with no live child cascades until the committed spine. The two compose: LCP commit publishes full winner pages into the APC index so a later session can hash-hit them (§3.4). Abort frees $b\ell_i$ bytes, not $b(L + \ell_i)$. TTL and offload are node-granular: speculative leaves first, then idle committed leaves, then the spine [20, 43].

3.3 Admission and Commit

A candidate of parent u is a pair $(x^k, \hat{x}^o)$ with admit probability p and prefix-hit rate q . Known suffixes ($p=q=1$) precede observation residuals ($q<1$), subject to $T_{\text{idle}}(u)$, residual HBM, and branch cap m . The object is an input prefix, not a draft output token [18, 7].

LCP commit. On prompt x at u ,

$$v^* = \arg \max_v \text{LCP}(x, x_v), \quad g_u \leftarrow g_u + 1. \quad (3)$$

Pages of the LCP become committed; the unmatched tail is a committed prefill; every sibling aborts. A job tagged

$g \neq g_u$ drops in $O(1)$. Committed decode attends only $\text{KV}(x_{v^*})$, so the emitted tokens match a baseline that prefills committed tokens only.

3.4 Cost-Ordered Prefill Pruning and Transfer Gating

Copy-on-write fixes trunk placement. It does not decide which residual enters a GPU kernel, nor which KV crosses the prefill–decode connector. **APP** makes both decisions per residual, (W_i, X_i) , by a cascade whose stage cost is strictly increasing (Figure 3). The stages are a block-hash probe, a CPU draft score, a prefix of the prefill kernel, the miss-tail prefill, and a connector `insert`. Stage $i+1$ runs only when stage i does not reject ρ_i . The committed winner is exempt: index 0 is forced into $\mathcal{K}$, and speculative pages stay unpublished until (??). CoW remains the storage substrate; **APP** is the filter on compute and on transfer. On the Qwen3-14B GSM8K forest, copy-on-write already stores less KV than APC, yet median fan-out is 140 ms against APC’s 87 ms: the k residuals are still submitted to prefill. The cascade is the stage that removes that work.

Action. Let H be the published radix index and P the page size. The matched prefix, the loop set, and the early-abort bit are

$$\begin{aligned} m_i &= \max\{jP : jP \leq \ell_i, \rho_i[0:jP) \in H\}, \\ \Lambda &= \{i \neq 0 : \text{Loop}(\rho_i)\}, \\ F_i &= [s(\rho_i[0:\lceil \alpha \ell_i \rceil]) < \tau]. \end{aligned} \quad (4)$$

The trunk is the alias $\sigma_i \leftarrow (\sigma_u, |x_u|)$, so $L \notin W_i$. With $\mathcal{A} = \{\text{HS}, \text{HP}, \text{DS}, \text{EA}, \text{PF}\}$ and first-match Iverson brackets,

$$\begin{aligned} a_i &= \text{HS}[m_i=\ell_i] + \text{HP}[0 < m_i < \ell_i] + \text{DS}[m_i=0][i \in \Lambda] \\ &\quad + \text{EA}[m_i=0][i \notin \Lambda][F_i] + \text{PF}[m_i=0][i \notin \Lambda][\neg F_i]. \end{aligned} \quad (5)$$

Index 0 is forced into $\mathcal{K}$, and $a_0 \in \{\text{HS}, \text{HP}, \text{PF}\}$.

$$W_i = [\alpha \ell_i][a_i=\text{EA}] + (\ell_i - m_i)[a_i=\text{HP}] + \ell_i[a_i=\text{PF}], \quad (6)$$

$$X_i = W_i[i \in \mathcal{K}][a_i \in \{\text{HP}, \text{PF}\}]. \quad (7)$$

?? is (5)–(7) on one radix tree: green pages lie in H , orange pages are grown, gray pages are withheld.

$$d_i^\times = D \cdot [i = 0 \vee s_i \geq \tau]. \quad (8)$$

(8) drops every $s_i < \tau$ before prefill. Algorithm 1 keeps DS only for $i \in \Lambda$; every other low score is prefilled and probed. Table 3 is the resulting first-expand span.

Lemma 1 (First-fan-out dominance). *On a cold index, $W_{\text{APP}} \leq W_{\text{CoW}} \leq W_{\text{APC}} = W_{\text{PD}}$ and $X_{\text{APP}} \leq X_{\text{PD}}$, with strict inequality in W whenever at least one repeated loop is draft-skipped, and strict inequality in X whenever $\mathcal{K} \neq [k]$ or any kept residual is a hash hit.*

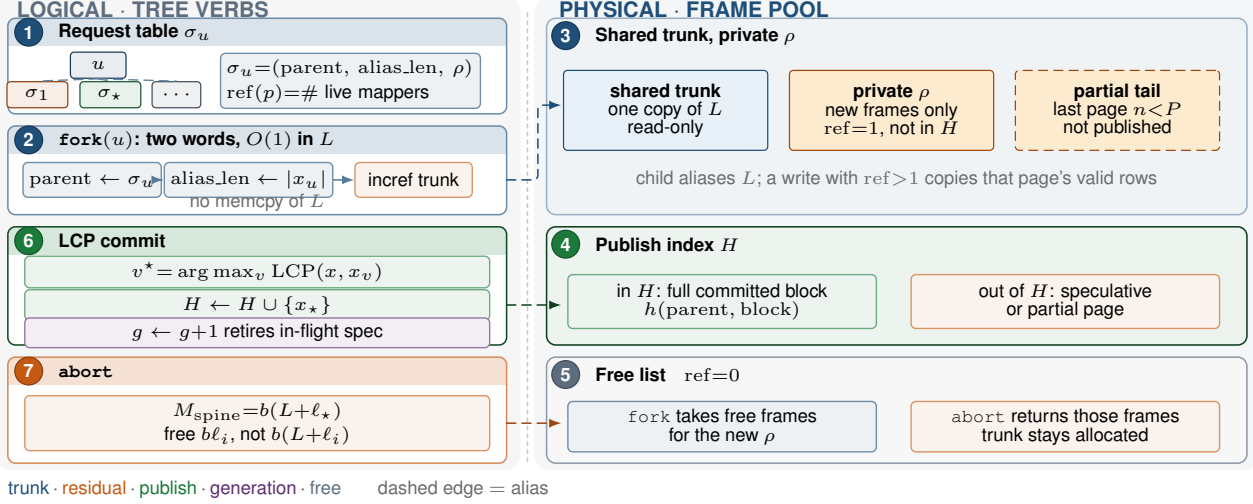


Figure 2. **Context-tree page pool.** `fork` aliases the trunk in $O(1)$ and allocates only ρ ①–③. LCP commit publishes x_* into H ④⑥. `abort` returns residual frames only, so live occupancy is the spine $M_{\text{spine}} = b(L + \ell_*)$ ⑤⑦.

Table 3. Work, transfer, and memory spans (bottleneck in red). W_{pre} is GPU prefill tokens; X is connector tokens; M is live KV after abort; T_{ab} is abort cost in pages; replay is a later `open` of the published winner. $\text{PAR}_{\text{trunk}}$ is the number of trunk copies touched on the first expand.

	W_{pre}	X	M after abort	T_{ab}	$\text{PAR}_{\text{trunk}}$	Replay W_{pre}
Recompute	$kL + \sum_i \ell_i$	0	$kL + \sum_i \ell_i$	$\Theta(kL/P)$	k	$kL + \ell_*$
APC (first expand)	$kL + \sum_i \ell_i$	0	$kL + \sum_i \ell_i$	LRU of hashed ρ	k	0
hash prefill (first)	$kL + \sum_i \ell_i$	0	$kL + \sum_i \ell_i$	LRU	k	0
P/D disagg.	$kL + \sum_i \ell_i$	$kL + \sum_i \ell_i$	$kL + \sum_i \ell_i$	full KV transfer	k	0
FS CoW	$L + \sum_i \ell_i$	0	$L + \ell_*$	$\Theta(\ell_i/P)$	1	L_{miss}
FS+ + P/D	$L + \sum_{i \in K} \ell_i$	$\sum_{i \in K} W_i$	$L + \ell_*$	lazy, residual only	1	0

$P=16$. Known-suffix items ($p=1$) are admitted before residuals ($p < 1$). Spec pages are never hashed; LCP commit publishes full blocks of x_* .

Hash, probe, and transfer. The index key is `hash(parent, block_tokens, extra)` [17]; only full pages enter H , so m_i steps by P , and $(?)$ is what later yields $m_i = \ell_i$. A hash-local hit is already APC-visible on the decoder, hence $X_i = 0$. On a miss, DS is only $i \in \Lambda$. Every other $s_i < \tau$ is prefilled, decoded for t_0 tokens, and extended iff $s'_i \geq \tau$ (Algorithm 1); that probe is not the EA edge. Default $\tau_{\text{pre}} = 0.15$; MATH-500 uses $\tau = 0.45$ and $t_0 \in \{64, 128\}$. Baseline disaggregation ships (1). APP ships (7).

$$T_{\text{fan}} = T_{\text{pre}}(W) + \mu X + T_{\text{CoW}} + T_{\text{abort}}. \quad (9)$$

T_{CoW} is a pointer swap. The blanket-skip model of §4.2 gives 139 ms (APC), 162 ms (baseline disaggregation), 47 ms (copy-on-write), and 31.4 ms. The GPU rule withholds only the loop and cuts computed prefill by 23% (§4.2).

3.5 The Spine on the Critical Path

After APP, fan-out is the prefill of the kept residual. End-to-end time is still $T_{\text{fan}} + T_{\text{dec}}$, and T_{dec} is the long term. Three placements put the spine on that sum. A fixed chunk width

and a fan-out barrier leave the freed KV and the pruned queue off the path.

Slots follow the spine. Let H be the KV pool after weights. A live tree occupies $M_{\text{spine}} = b(L + \ell_*)$. APC still holds the aborted residuals until they expire, so its live set is the larger M_{APC} . The number of concurrent decode slots is

$$C = \lfloor H/M \rfloor, \quad (10)$$

with $M = M_{\text{spine}}$ for ForkServe and $M = M_{\text{APC}}$ for the prefix cache. C is not a constant chunk size. A request that finishes its current decode step releases the slot, and the next queued spine takes it. Throughput is

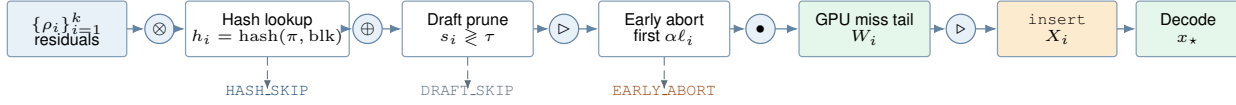
$$\Theta \propto C/T_{\text{e2e}}. \quad (11)$$

The same H therefore admits more sessions once occupancy is the spine, and the wall clock of a batch is the number of waves n/C , not n divided by a fixed width.

Prefill the next winner during decode. Each tick t has a token budget B_t from the chunked-prefill cap [2]. JIT-

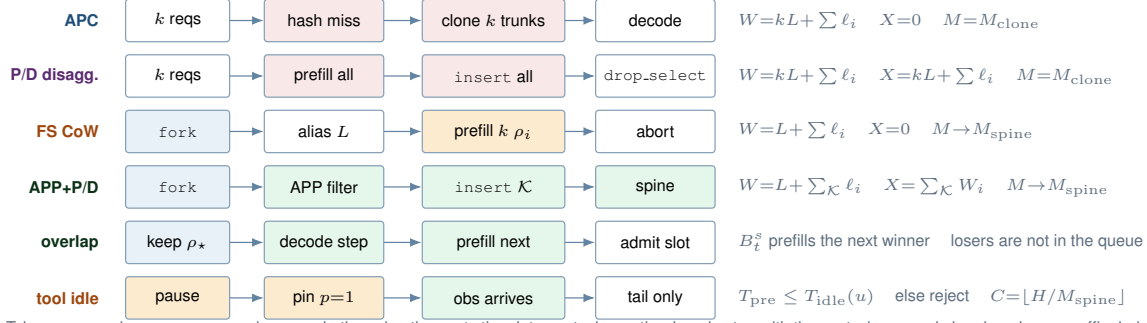
ρ_i : i -th residual h_i : block hash s_i : draft score α : early-abort fraction κ : chunk W_i : GPU miss tail X_i : connector tokens a_i : action

(a) APP + P/D pipeline (Alg. 1)



Parallelism: k residuals Layout: location-addressed CoW, not content-addressed APC Winner $i=0$ is never dropped Spec pages unpublished
Work $W = \sum_i W_i$ Transfer $X = \sum_{i \in \mathcal{K}} W_i$ Occupancy after abort $M_{\text{spine}} = b(L + \ell_*)$ Replay of x_* : $W = X = 0$

(b) First-fan-out dataflow (same k, L, ℓ). Bottleneck in red.



Takeaway: pruning removes non-winner work; the spine then sets the slot count, shares the decode step with the next winner, and absorbs a known suffix during the tool pause.

Figure 3. Cost-ordered prefill pruning and the critical path. (a) A residual exits at the first decisive stage. Only a kept miss tail reaches insert. (b) APC and baseline disaggregation clone the trunk. CoW aliases L . APP keeps the winner. The overlap row prefills that next winner in the decode step. The tool-idle row pins a known suffix during the pause and leaves the observation tail.

Serve’s margin [52] gives each committed request r a minimum token rate $\underline{s}_r$.

$$B_t^c = \min\left(B_t, \sum_{r \in R_t^c} \max(\underline{s}_r \Delta, 1)\right), \quad (12)$$

$$B_t^s = B_t - B_t^c. \quad (13)$$

B_t^c is committed decode. B_t^s is not spent on the $k-1$ residuals APP already rejected. It prefills the miss tail of the next winner, in the same step as the running decode. If $B_t^s = 0$, no speculative chunk is admitted. The fan-out of losers is not a generate that must finish before any decode token. A tree-sticky router pins the trunk to the worker that prefills the root; only residual work may migrate.

Hide the known suffix in the tool pause. A candidate with $p=1$ is an exact wrapper, not a guess. If $T_{\text{pre}}(|w|) \leq T_{\text{idle}}(u)$, that prefill runs while the tool is in flight, including when the parent is still marked committed. The pause is the deadline. A suffix that would finish after $T_{\text{idle}}(u)$ is rejected, as is any observation residual the pause cannot cover. When the observation arrives, LCP commit checks the pinned prefix and the committed prefill is the unmatched tail. Under P/D the same suffix is prefiller work tagged soft-deadline = $T_{\text{idle}}(u)$ and is inserted once, onto the trunk owner. Speculative tokens still do not enter the sampler, as under (??).

4 EVALUATION

The GPU measurements share one node and one engine, and they differ by model, tensor-parallel width, and workload. §4.1 is the full GSM8K forest on Qwen3-14B at $\text{tp}=2$. §4.2 is the earlier blanket-skip cost model of APP, with no GPU, against APC, hash prefill, baseline prefill–decode disaggregation, ForkServe, and three decoding-time pruners; §4.2 replaces that skip. §4.2 measures the GSM8K prefill threshold $\tau_{\text{pre}}=0.15$ on Qwen3-8B and the accuracy–latency trade-off when that rule sits in front of ESC, Speculative Rejection, and DPTS. §4.3 runs ForkServe+ on Qwen3-8B at $\text{tp}=2$ and $\text{tp}=4$. §4.4 is a three-turn Tree-of-Thoughts on Qwen3-4B and Qwen3-8B, one problem per generate. §4.5 repeats the forest on six families at $n=40$. §4.6 sweeps branching, depth, pruning, and batch width on Llama-3-8B, Mistral-7B, and DeepSeek-R1-Distill-Llama-8B, then runs a 2048-token contest forest on that DeepSeek model and on Qwen3-4B.

Setup. GPU runs use vLLM on four A100-SXM4 40GB GPUs, with `bfloat16` weights and $\text{tp}=2$ or 4. The models are Qwen3-14B, Qwen3-8B, Qwen3-4B, Qwen2.5-Math-1.5B, Llama-3-8B, Mistral-7B, and DeepSeek-R1-Distill-Llama-8B. Tree-of-Thoughts uses $k=4$ and a 512-token budget (2048 on the contest forest), in batches of eight. The three-turn run in §4.4 issues one problem per generate. A

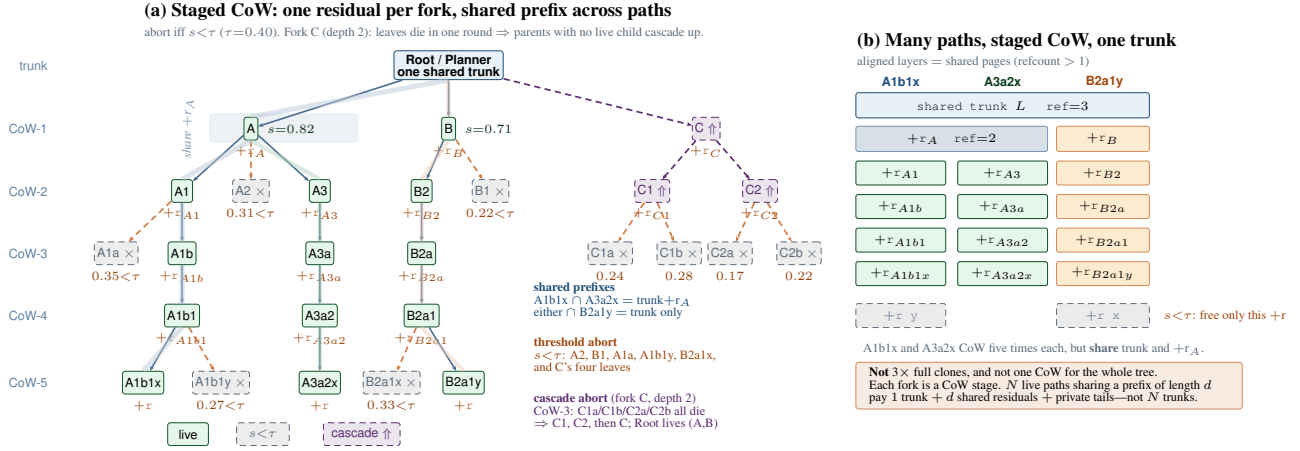


Figure 4. Staged CoW along many paths, threshold abort, cascade abort, shared prefixes. (a) Each edge is one CoW: the child aliases the parent’s pages and allocates only a residual ($+r$). Live paths A1b1x, A3a2x, and B2a1y share the trunk; the first two also share $+r_A$. Fork C has depth 2: at CoW-3 all four leaves die, so abort cascades through C1 and C2, then C; it stops at Root, which still has live children A and B. (b) KV stacks of the three live leaves, layers aligned: a shared layer is one physical page set with refcount > 1 , not N copies. Aborting A1b1y or B2a1x frees only that leaf residual; cascade-aborting C also frees $+r_{C1}$, $+r_{C2}$, and $+r_C$.

Algorithm 1 Corrected admission, replacing (8). Λ is the repeated-loop set. y_i is a probe of t_0 tokens and s'_i is its draft score. Iverson brackets select the decode length. Only Λ keeps the old skip.

Input: $\{\rho_i\}_{i=0}^{k-1}$, threshold τ , probe t_0 , budget D
Output: action a_i , decode length d_i , set $\mathcal{K}$

```

1:  $\Lambda \leftarrow \{i \neq 0 : \text{Loop}(\rho_i)\}$ 
2: for  $i = 0$  to  $k - 1$  do
3:    $s_i \leftarrow \text{DraftScore}(\rho_i)$ 
4:   if  $i \in \Lambda$  then
5:      $(a_i, d_i) \leftarrow (\text{DS}, 0)$  {was  $d_i^x = 0$  for all  $s_i < \tau$ }
6:   else if  $i = 0 \vee s_i \geq \tau$  then
7:      $(a_i, d_i) \leftarrow (\text{PF}, D)$ 
8:   else
9:      $y_i \leftarrow \text{Dec}(x_u || \rho_i, t_0)$ 
10:     $s'_i \leftarrow \text{DraftScore}(y_i)$ 
11:     $(a_i, d_i) \leftarrow (\text{PF}, D[s'_i \geq \tau] + t_0[s'_i < \tau])$ 
12:   end if
13: end for
14:  $\mathcal{K} \leftarrow \{i : a_i \neq \text{DS}\}$ 
15:  $W_i \leftarrow \ell_i \cdot [a_i = \text{PF}]$ 
16:  $\text{DecTokens} \leftarrow \sum_{i=0}^{k-1} d_i$ 

```

math answer is correct when its final number matches the reference.

4.1 GPU forest

Setup. This section uses Qwen3-14B at $\text{tp}=2$. The GSM8K test set has 1319 problems. Each problem is expanded with four thoughts and decoded for at most 512 tokens, in 165 batches of eight. Accuracy is the fraction of problems whose final number matches the reference.

Table 4. GSM8K test, Qwen3-14B, $\text{tp}=2$, $n=1319$. Median over 165 batches of eight. Best storage in green.

	recompute	APC	FS
Accuracy	0.847	0.844	0.845
Peak KV, median (tok)	4136	1430	1030
Peak KV, max (tok)	4944	1632	1232
Peak / APC	$2.89 \times$	$1.00 \times$	$0.72 \times$
Fan-out, median (ms)	326	87	140
Generation (min)	—	18.6	18.9
KV saving vs. rec.	—	0.65	0.75

Storage and expand latency. Table 4 reports the storage result and the expand-latency gap that APP is designed to close. Median peak KV is $0.72 \times$ APC and $0.25 \times$ recompute, and accuracy stays within one point of both. Storage and fan-out do not share that ordering: median fan-out is 140 ms for ForkServe and 87 ms for APC, because copy-on-write and abort execute on the expand path, whereas an APC hash hit does not. End-to-end generation remains within 2% of APC (18.9 vs. 18.6 minutes). The same peak ordering, $0.58\text{--}0.80 \times$ APC, holds on seven math sets for Qwen3-14B, Qwen3-8B, and DeepSeek-R1-Distill-Llama-8B at $\text{tp}=2$ and $\text{tp}=4$. The peak is unchanged when the tensor width doubles; it follows the committed path, as (2) requires.

4.2 Prefill pruning

Setup. This subsection is the earlier blanket-skip cost model, not the GPU rule of §???. Control plane, no GPU. Ten sessions, $k=4$, $L=256$, $\ell=32$. Two of four residuals fail the draft score and are withheld before prefill; the winner is index 0. Prefill time is $\ell \times 12 \mu\text{s}/\text{token}$; P/D transfer

in H
 withheld
 one box = P tokens, shared radix root

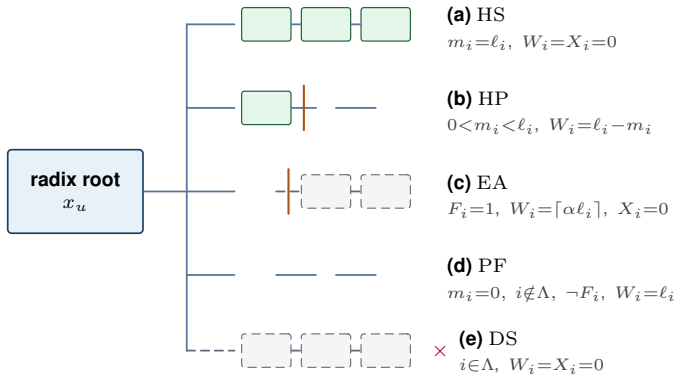


Figure 5. **One radix tree, five actions** ((5)–(7)). The root is the aliased trunk x_u . Each edge is one page of P tokens. Green pages are in H ; orange pages are prefilled; gray pages are withheld. (a) HS: $m_i = \ell_i$, $W_i = X_i = 0$. (b) HP: $0 < m_i < \ell_i$, $W_i = \ell_i - m_i$. (c) EA: $m_i = 0$, $F_i = 1$, $W_i = \lceil \alpha \ell_i \rceil$, $X_i = 0$. (d) PF: $m_i = 0$, $i \notin \Lambda$, $-F_i$, $W_i = \ell_i$. (e) DS: $i \in \Lambda$, $W_i = X_i = 0$. A low score with $i \notin \Lambda$ is not (e); it is prefilled and probed (Algorithm 1).

Table 5. Control-plane fan-out (10 sessions, $k=4$, $L=256$, $\ell=32$). Bottleneck in red; ForkServe+ in green. Not GPU wall-clock.

Phase	Method	W	T_{fan}	X	peak	skip	prune
fan-out	APC	11520	139.0	0	11520	0	0
	hash	11520	139.0	0	11520	0	0
	P/D disagg.	11520	162.1	11520	11520	0	0
	FS	3840	47.1	0	3840	0	0
	ForkServe+	2592	31.4	32	2880	9	30
	ForkServe+ / APC	-77.5%	-77.4%	—	-75%	—	—

is $2 \mu\text{s}/\text{token}$. APC and hash prefill publish blocks only after a request materializes them, so the first fan-out clones k trunks. Baseline disaggregation uses that prefill and then transfers the live token span. ForkServe aliases the trunk by pointer swap and prefills the k residuals. APP adds hash skip, draft prune, early abort, lazy abort, and a connector that inserts only retained residuals.

Fan-out. Table 5 is the comparison the GPU gap motivates. On the first expand, hash prefill matches APC: the blocks are unpublished, so the hash probe does not remove the cloned trunks. Baseline disaggregation exceeds APC (162 vs. 139 ms) because it transfers that clone. ForkServe removes the clone by aliasing the trunk (3840 prefill tokens, 47 ms). APP then withholds prefill from 30 of the 40 residuals, which reduces T_{fan} to 31.4 ms and the connector payload from 11520 tokens to 32.

Decoding-time pruning. Table 6 runs ESC [21], Speculative Rejection [31], and DPTS [19] on this same fan-out. Each method prefills the shared trunk once. The length of the decode that produces the prune score is the length published for that method. ESC stops after a window of 5

Table 6. Decoding-time pruning on the Table 5 fan-out (10×4 , $L=256$). Decode-to-cut is the prefix that must be generated before a child can be dropped. “On losers” counts that prefix on the two low-score children. Same $12 \mu\text{s}/\text{token}$ clock. Not GPU wall-clock.

	prefill	decode	on losers	peak	T_{cut}
ESC	2560	20480	10240	23040	276.5
Spec. Rejection	2560	10240	5120	12800	153.6
DPTS	2560	4000	2000	6560	78.7
ForkServe+	2592	0	0	2880	31.4

agreeing samples; this fan-out has four children and two low-score answers, so the window stays open and every child is decoded to the 512-token budget. Speculative Rejection is charged its partial-reward horizon $\tau=256$ and then drops the lower half ($\alpha=0.5$); the memory-limit trigger in that paper would not fire at $k=4$, so τ is the earlier checkpoint. DPTS early-stops after its 100-token mini-step. The two low-score children are the ones removed, and the winner is kept. Decoded tokens use the same $12 \mu\text{s}/\text{token}$ clock. DPTS is the earliest cut of the three and still decodes 4000 tokens before the drop, 2000 of them on losers, reaching the cut at 78.7 ms. APP has already withheld those residuals: 0 tokens on losers and $T_{\text{fan}}=31.4$ ms. That row is the blanket skip. §4.2 replaces it: only a repeated loop is withheld before prefill.

Probe-then-grow. The blanket skip is the wrong cut once the residual is not a loop. We measure the rule in Algorithm 1 on Qwen3-8B, one GPU each, four A100-40GB, $\text{tp}=1$. The slice is the first 32 MATH-500 problems at level 4 or 5. Temperature is 0, EOS is ignored, prefix caching is on, and the decode budget is $D=1024$. Half of the k

Table 7. MATH-500 level ≥ 4 , Qwen3-8B, $n=32$, $D=1024$. Any-finished is exact match on a branch that ran the full budget. “Drop $s < \tau$ ” withholds every low-score residual before prefill. “Grow t_0 ” is Algorithm 1: loops stay at $d_i=0$; every other low-score residual is probed for t_0 tokens and continues to D only if $s'_i \geq \tau=0.45$. Time is trunk warm plus fan-out, in seconds.

	$k=4$	$k=8$	$k=16$
Full fan-out	12/32, 25.3	15/32, 52.5	20/32, 104.1
Drop $s < \tau$	6/32, 19.1	11/32, 25.6	13/32, 52.1
Skip loops only	11/32, 23.0	17/32, 41.5	16/32, 71.4
Grow $t_0=64$	8/32, 24.0	16/32, 42.0	18/32, 72.5
Grow $t_0=128$	9/32, 24.7	13/32, 43.3	20/32, 72.2

Table 8. GSM8K, Qwen3-8B, $n=16$, $D=512$, ESC only. Any-finished is exact match on a full-budget branch. $\tau_{\text{pre}}=0.15$ drops the loop; 0.45 also drops illegal text. Time is trunk warm plus fan-out, in seconds. Computed prefill falls 23% at 0.15 and 45% at 0.45 at every k .

	$k=4$	$k=8$	$k=16$
ESC	16/16, 8.65	16/16, 10.84	16/16, 17.05
APP $\tau_{\text{pre}}=0.15$	15/16, 8.22	16/16, 10.19	16/16, 14.14
APP $\tau_{\text{pre}}=0.45$	14/16, 7.89	14/16, 8.60	16/16, 10.93

residuals are contest strategies. The other half alternates a repeated loop with illegal text. A question counts when some branch that ran all 1024 tokens matches the gold answer. The winner is residual 0.

Table 7 is the correction. Dropping every residual below τ cuts any-finished from 12/32 to 6/32 at $k=4$ and from 20/32 to 13/32 at $k=16$. Loop branches are 0 for 0 at every k , so skipping them before prefill is safe. At $k=16$, a 128-token probe extends every illegal residual (128/128) and none of the loops. Any-finished returns to the full fan-out, 20/32, while end-to-end time falls from 104.1 s to 72.2 s and decode tokens from 524288 to 393216. Probing the loops as well extends none of them and costs 74.4 s for the same 20/32. At $k=8$ the shorter probe is the one that holds accuracy: grow at $P=64$ is 16/32 in 42.0 s, against 15/32 in 52.5 s for the full fan-out. At $k=4$ the probe does not recover the full fan-out (9/32 against 12/32); skipping the single loop, and decoding the illegal residual to D , is 11/32 in 23.0 s.

Accuracy-speed trade-off. The MATH-500 probe still uses a decode threshold $\tau=0.45$ on the generated prefix. Prefill admission is a different score, with default $\tau_{\text{pre}}=0.15$: it drops only the repeated loop (0.02) and starts illegal text (0.22) and every live strategy (1.0). We measure that rule on Qwen3-8B GSM8K, $n=16$, $D=512$, the same hopeless mix, four A100-40GB, $\text{tp}=1$. The host decoder is ESC, so a withheld child never pays residual prefill or a decode slot. A denser sweep of τ_{pre} collapses to three admission sets; 0.15 and 0.45 are the two nonzero operating points.

Table 8 is the trade-off. $\tau_{\text{pre}}=0.15$ keeps 15/16, 16/16,

Table 9. Same GSM8K slice, $k=16$, $\tau_{\text{pre}}=0.15$ in front of three host decoders. Admitted children and computed prefill tokens are identical across hosts: prefill admission does not read τ , d_{DPTS} , or α . Any-finished stays 16/16.

	adm.	prefill	e2e (s)	any
ESC	16	5008	17.05	16/16
ESC+APP	12	3848	14.14	16/16
Spec. Rejection	16	5008	13.14	16/16
SR+APP	12	3848	12.47	16/16
DPTS	16	5008	11.86	16/16
DPTS+APP	12	3848	11.49	16/16

16/16 and still cuts wall-clock time, more as the tree widens (−5%, −6%, −17%). Computed prefill tokens fall from 1320/2624/5008 to 1018/2020/3848. Raising τ_{pre} to 0.45 is the extra-speed point: −9%, −21%, −36% end-to-end, and 732/1448/2768 computed prefill tokens. It drops two questions at $k=4$ and $k=8$. At $k=16$ both thresholds stay at 16/16, so 0.45 is the faster matched-accuracy setting (10.93 s vs. 14.14 s). The default is 0.15 because it holds accuracy at every k .

Table 9 stacks the same $\tau_{\text{pre}}=0.15$ in front of each host. ESC, Speculative Rejection, and DPTS alone all report 5008 computed prefill tokens: a decode-time cut still prefills every child. Adding APP admits 12 of 16 residuals and 3848 prefill tokens on every host. Wall-clock falls 17% on ESC, 5% on Speculative Rejection, and 3% on DPTS, because those last two hosts would have stopped the loop later anyway. The accuracy that Table 8 preserves is unchanged: every row in Table 9 is 16/16. At $k=8$ the same threshold is 16/16 on ESC (10.84 to 10.19 s) and lifts Speculative Rejection and DPTS from 15/16 to 16/16.

Concurrency from measured KV. Our empirical results on DeepSeek-R1 GSM8K (761 APC tokens, 561 ForkServe tokens, $n=16$, $k=4$) in an 8 GiB pool in (11) give 76 concurrent trees for APC and 103 for ForkServe. Figure 6 sweeps offered QPS: under a 1 s P99 TTFT constraint the same model admits 384 QPS for APC and 512 for the pruned tree, and saturated decode throughput rises from 48640 to 65920 tokens/s (+35.5%). This uses measured M , not the control-plane peak of Table 5.

4.3 GPU ForkServe+

Setup. Qwen3-8B, bfloat16 , $n=80$, $D=512$, $\text{tp}=2$ and 4 on four A100-SXM4 40GB GPUs. ForkServe+ enables lazy abort, pointer swap, hash skip, draft prune, and the P/D gate. GSM8K and Game24 are ToT ($k=4$); HumanEval is ReAct wrap+recovery ($k=2$).

Peak KV on this forest is 1182 against APC 1582 ($0.75\times$) at both widths. Game24 and HumanEval peaks are $0.62\times$ and $0.96\times$ APC.

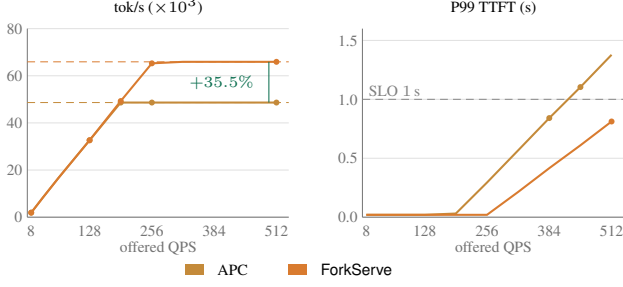


Figure 6. **SLO sweep** from measured DeepSeek GSM8K peaks. Left: saturated tok/s plateaus at $C \times$ decode-rate (76 vs. 103 slots). Right: APC crosses the 1 s P99 line at 448 QPS; ForkServe stays under it through 512.

Table 10. Qwen3-8B GPU forest, $n=80$, $D=512$. Peak KV is independent of tp . Game24 / HumanEval peaks stay with the spine.

	peak	D^+	e2e	acc.	time reduction	G24	HE
APC $tp=2$	1582	40960	36.0	0.900	—	1289	1970
ForkServe+ $tp=2$	1182	13356	28.2	0.863	0.78 $\times$	793	1882
APC $tp=4$	1582	40960	22.7	0.938	—	1289	1970
ForkServe+ $tp=4$	1182	12826	16.9	0.925	0.75 $\times$	793	1882

4.4 Multi-turn Tree-of-Thoughts

Setup. Qwen3-4B and Qwen3-8B, $bfloat16$, $tp=2$, temperature 0, $n=16$. Each problem is one session of $k=4$ agents and three turns, and each system issues that problem as its own generate. Turns 1 and 2 decode a 32-token step; the losers abort; the winner spine is the parent of the next fan-out. Turn 3 is the published answer, $D=512$ on GSM8K and $D=256$ on Game24.

Depth keeps one spine. Table 11 and Figure 8 are GSM8K and Game24 under that schedule. Peak KV is 275 against APC 318 on GSM8K (0.86 $\times$) and 216 against 265 on Game24 (0.82 $\times$), on both models. ForkServe stays within 6% of APC end-to-end (1.00–1.06 $\times$). Across 16 problems, three turns, and four agents, the expand issues 192 residuals. ForkServe+ prefills 48 of them and withholds the other 144 before they reach the GPU.

Table 12 and Figure 9 separate that expand from the winner decode. The expand speedup is 5.0–6.4 $\times$ for ForkServe and 7.9–10.6 $\times$ for ForkServe+. Decode does not match it. ForkServe decode is slower than APC because the two step decodes are timed there, which is why end-to-end stays at 1.00–1.06 $\times$. GSM8K accuracy matches APC on Qwen3-4B (15/16 on the same problems) and is one problem lower on Qwen3-8B (15/16 vs. 16/16). On Game24, ForkServe solves two more problems than APC on Qwen3-4B (8/16 vs. 6/16) and the same nine on Qwen3-8B; ForkServe+ solves 9/16 and 10/16.

Table 11. Three-turn ToT, $n=16$, $k=4$, $tp=2$, one problem per generate. End-to-end time is seconds per problem. Peak KV is tokens on the live spine. FS is ForkServe; FS+ is ForkServe+. Shortest end-to-end time and lowest peak in green.

	Qwen3-4B			Qwen3-8B		
	APC	FS	FS+	APC	FS	FS+
GSM8K acc.	0.938	0.938	0.938	1.000	0.938	0.938
GSM8K e2e (s)	1.66	1.76	1.63	2.48	2.56	1.91
GSM8K peak	318	275	275	318	275	275
Game24 acc.	0.375	0.500	0.562	0.562	0.562	0.625
Game24 e2e (s)	1.61	1.64	1.13	2.15	2.15	1.52
Game24 peak	265	216	216	265	216	216

Table 12. Expand versus decode, milliseconds per problem. Expand is the prefill of the k residuals across three turns. On APC that timer also includes the two 32-token step decodes; ForkServe and ForkServe+ charge those steps to decode. Speedup in Figure 9 is APC expand time over the other system. Shortest time in each trio in green.

	Expand (ms)			Decode (ms)		
	APC	FS	FS+	APC	FS	FS+
4B GSM8K	429	83	55	1215	1680	1572
4B Game24	435	87	53	1166	1552	1074
8B GSM8K	597	94	58	1867	2468	1848
8B Game24	598	94	56	1538	2060	1466

4.5 Cross-model forest

Setup. Same four systems, $n=40$, $D=512$, $tp=2$ and 4, on Qwen3-4B, Qwen2.5-Math-1.5B, Qwen3-14B, Llama-3-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Llama-8B.

Peak occupancy. APC retains the shared trunk and the union of the residuals, which is the copy-on-write term $M_{APC} = M_{CoW} = b(L + \sum_i \ell_i)$ from (2). The tree retains only the committed spine $M_{spine} = b(L + \ell_*)$. The ratio of these two terms cancels b and equals $(L + \ell_*) / (L + \sum_i \ell_i)$, so it depends on k and the lengths (L, ℓ_*) rather than on the weights or the tensor width. When the residuals have similar length, the ratio simplifies to $(L + \ell) / (L + k\ell)$ and decreases as L/ℓ decreases. Figure 7 shows this pattern: a long trunk on GSM8K gives 0.72–0.75, and a shorter trunk on Game24 gives 0.57–0.61. Figure 10 repeats the comparison at $n=80$ with recompute and ForkServe+ on separate axes. Relative to recompute, which stores the cloned occupancy M_{clone} , the spine is 0.25 $\times$. ForkServe+ matches that ratio. Accuracy remains at the APC level, which indicates that the lower peak is a storage effect on the same committed answer. Figure 11 shows the per-item decode-length distribution on GSM8K and Game24.

HumanEval uses $k=2$, so the residual is a smaller fraction of the context and M_{spine} remains close to M_{APC} . Pass@1 is unchanged across recompute, APC, and ForkServe.

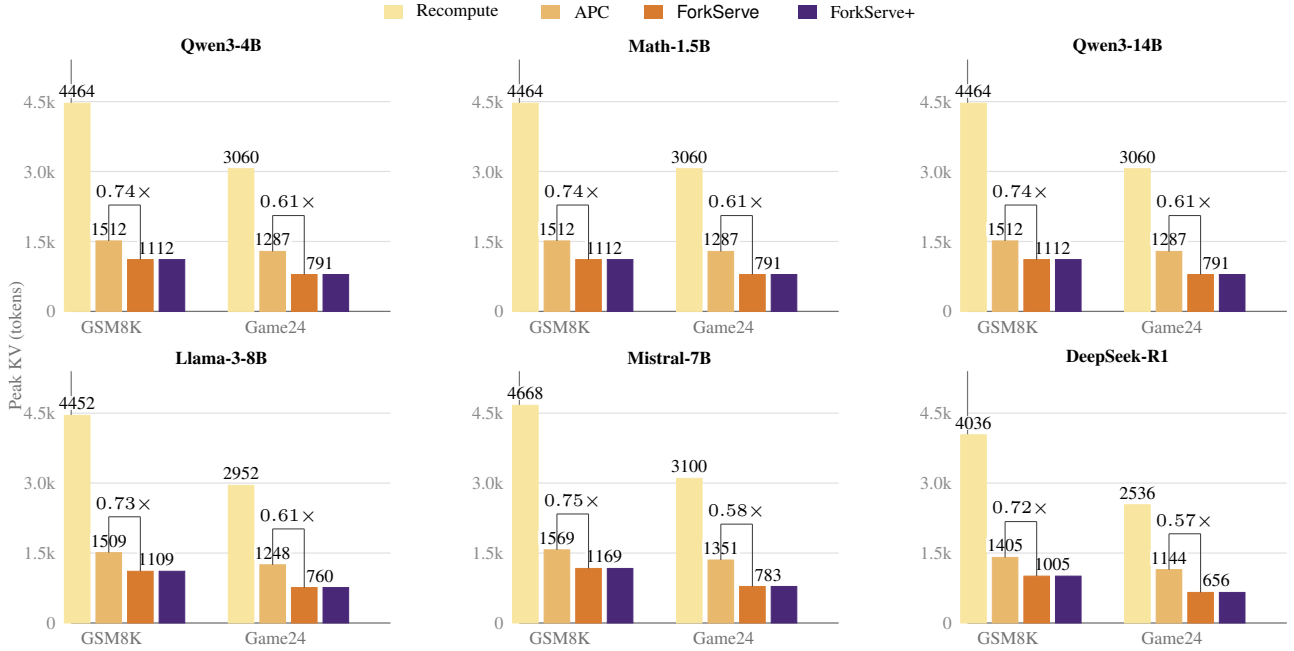


Figure 7. **Peak KV** on six models ($n=40$, $tp=2$). Each panel is one family; groups are GSM8K and Game24. Bars are recompute, APC, ForkServe, and ForkServe+ (cream / tan / orange / purple). ForkServe and ForkServe+ share the spine. Annotated $\times$ is $M_{\text{spine}}/M_{\text{APC}}$: 0.72–0.75 $\times$ on GSM8K and 0.57–0.61 $\times$ on Game24, independent of checkpoint size, as (2) predicts.

Table 13. Last-line answer stop, $n=32$, $tp=2$. ForkServe+ stops only on a finished answer in the last non-empty line. Peak is the max over chunks. GSM8K $D=512$; Game24 $D=256$. R1-8B is DeepSeek-R1-Distill-Llama-8B; L8B is Llama-3-8B; M7B is Mistral-7B. Best storage in green.

Model	Metric	APC	FS+
R1-8B	GSM8K peak	1493	1093
	GSM8K acc.	0.500	0.562
	GSM8K D^+	16384	12693
	Game24 peak	1064	616
L8B	GSM8K peak	1485	1085
	GSM8K acc.	0.719	0.688
	GSM8K D^+	16384	5101
	Game24 peak	1224	736
M7B	GSM8K peak	1521	1121
	GSM8K acc.	0.344	0.188
	GSM8K D^+	16384	6723
	Game24 peak	1296	728

4.6 Branching, pruning, and long traces

Setup. GSM8K, $n=16$, $D=512$, $tp=2$, on Llama-3-8B, Mistral-7B, and DeepSeek-R1-8B. The sweep changes k or adds a second ToT level. MATH-500 ($n=200$) and AIME ($n=73$) use $D=2048$.

The ratio falls with k . Table 14 and Figure 12 measures $(L+\ell)/(L+k\ell)$ on the GPU. The spine peak is invariant in k while the APC peak increases, so the ratio moves from 0.89–0.90 at $k=2$ to 0.53–0.55 at $k=8$ on all three families. At

Table 14. GSM8K peak KV versus branching ($n=16$, $tp=2$, $D=512$). The spine peak is invariant in k ; the APC peak increases with k . ratio is FS/APC. L8B is Llama-3-8B; M7B is Mistral-7B; R1-8B is DeepSeek-R1-Distill-Llama-8B. Best storage in green.

Model/ratio	Method	$k=2$	$k=4$	$k=8$	depth 2
L8B	APC	610	750	998	814
	FS	550	550	550	614
FS/APC		0.90	0.73	0.55	0.75
M7B	APC	629	769	1033	841
	FS	569	569	569	641
FS/APC		0.90	0.74	0.55	0.76
R1-8B	APC	629	761	1053	841
	FS	561	561	561	641
FS/APC		0.89	0.74	0.53	0.76

depth 2 the ratio is 0.75–0.76. ForkServe+ leaves the peak unchanged. At $k=8$ it rejects 28 branches and reduces fan-out from 307 to 179 ms on Llama-3-8B, from 349 to 191 ms on Mistral-7B, and from 363 to 194 ms on DeepSeek-R1. With draft pruning disabled, Llama-3-8B fan-out is 247 ms and DeepSeek-R1 fan-out is 271 ms. A batch of 16 keeps the ratio at 0.72–0.73 (2071/2871 on Llama-3-8B, 2135/2935 on Mistral-7B, 2093/2893 on DeepSeek-R1).

At $D=2048$ the ratio is 0.75–0.91 (Table 15). On MATH-500, ForkServe matches APC (432 s vs. 429 s on DeepSeek-R1). ForkServe+ cuts DeepSeek-R1 from 429 s to 247 s and Qwen3-4B from 329 s to 236 s; the Qwen3-4B $n=32$ slice is 38 s versus 53 s. ForkServe+ cuts MATH-500 fan-out

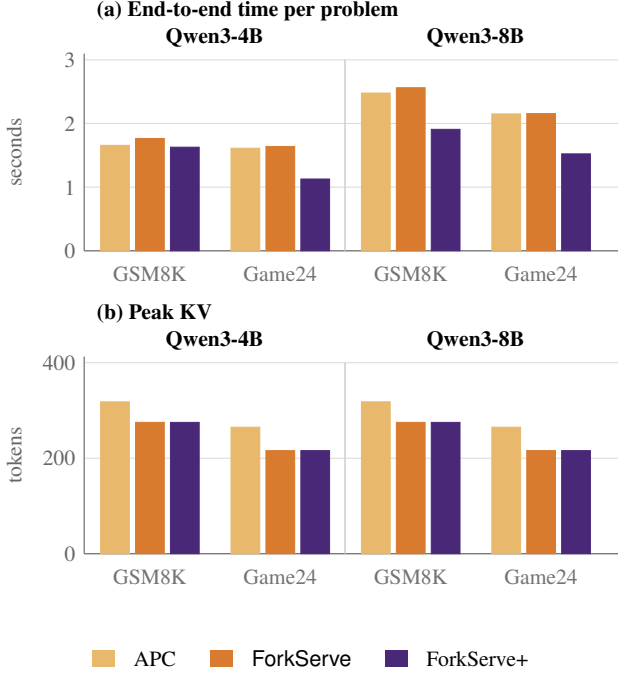


Figure 8. **Three-turn ToT** on Qwen3-4B and Qwen3-8B ($n=16$, $tp=2$, one problem per generate). (a) End-to-end seconds per problem; a shorter bar is faster. (b) Peak KV tokens. ForkServe stays with APC on latency and stores the spine ($0.86\times$ APC on GSM8K, $0.82\times$ on Game24). ForkServe+ is faster because it prefills one of four residuals.

from 2406 to 1284 ms on DeepSeek-R1 and from 2787 to 1014 ms on Qwen3-4B, against APC at 1593 and 1201 ms. Qwen3-4B ForkServe+ scores 120/200 on MATH-500 (APC 124/200).

5 RELATED WORK

Request-level serving. Orca [47] introduced iteration-level scheduling; vLLM [17] paged the KV cache and hashes full blocks for APC; SGLang [53] added radix prefix sharing; Sarathi [2, 3] chunked prefills; DistServe and Splitwise [54, 32] disaggregated prefill from decode. Prefix caching discovers sharing after tokens exist. Prefill–decode disaggregation isolates TTFT from inter-token latency; it does not select which branch to compute, and it transfers every page the prefiler produced. APP uses both mechanisms as filters: the hash as a skip, the connector as a gate.

Program- and SLO-aware scheduling. Autellix [25] schedules by program-level attained service; JITServe [52] allocates just-enough bandwidth under heterogeneous SLOs. Hermes [24], ThunderAgent [16], and SAGA [13] raise the atom to the program or workflow. These systems leave KV

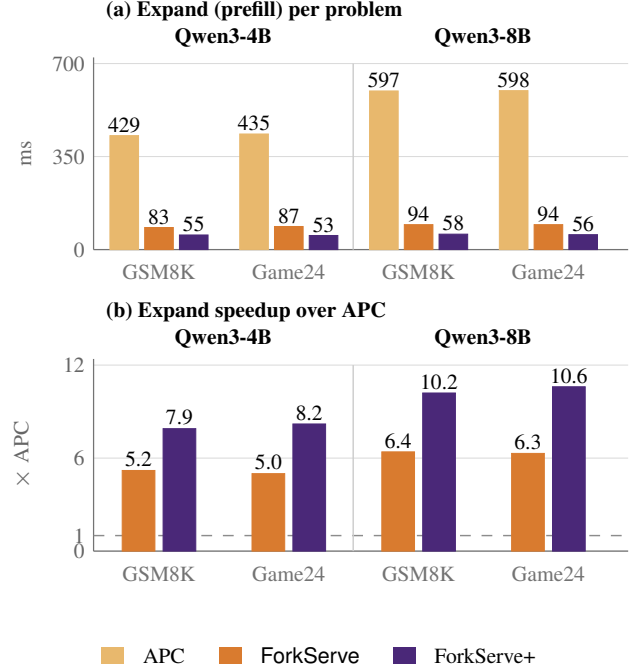


Figure 9. **Expand versus APC** on the same four runs as Table 12. (a) Expand milliseconds per problem. (b) Speedup of that expand over APC. The dashed line is $1\times$. ForkServe is $5.0\text{--}6.4\times$ and ForkServe+ is $7.9\text{--}10.6\times$. Decode in Table 12 does not repeat that ratio: the step decodes sit in the ForkServe decode column, so ForkServe end-to-end stays with APC.

unforked and unissued children unprefilled.

KV retention and overlap. InferCept [1], Continuum [20], MORI [43], and ConServe [11] keep the *current* context alive or placed. Sutradhara [6] overlaps tool execution with prefill of one linear tool-independent prefix. SmoothAgent [30] lookahead-transforms context; IdleSpec [9] drafts plans for accuracy, not KV. Classical speculative decoding [18, 7, 22] drafts *output* tokens. ForkKV [38] CoW-splits LoRA residuals. We represent the future as a *set* of CoW children, speculate several known suffixes, commit by LCP, prune losers before prefill, and gate P/D insert.

Decoding-time pruning. A second line drops a candidate after the model has generated it. Beam search in Tree-of-Thoughts [45] scores a finished step and keeps the top beams. Early-stopping self-consistency (ESC) [21] draws a window of complete samples and skips later samples only once that window agrees. Speculative Rejection [31] ranks partial generations and halts the lower quantile. Dynamic Parallel Tree Search (DPTS) [19] generates a fixed mini-step on every active node, then early-stops low-confidence children. In each method the score is computed from tokens already

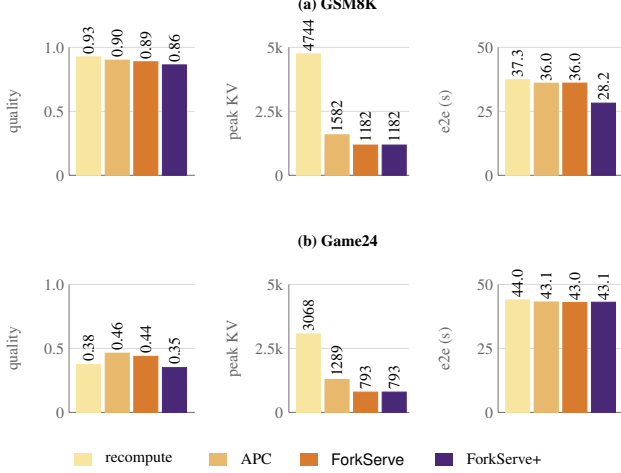


Figure 10. Four systems on Qwen3-8B ($n=80$, $tp=2$, $D=512$). Each panel has its own axis. Quality is higher-better; peak KV and end-to-end time are the measured values, so a shorter bar is better. (a) GSM8K. (b) Game24. Figure 10 shows that ForkServe and ForkServe+ lower peak KV below APC on both workloads ($0.75\times$ on GSM8K, $0.62\times$ on Game24) while GSM8K quality stays with APC, and that ForkServe+ also lowers GSM8K end-to-end time from 36.0 s to 28.2 s.

resident in the KV cache, so the prefill of the candidate and a decode prefix are issued before the drop. APP scores a known residual first. A repeated loop is withheld before prefill, which also keeps that child off the disaggregated connector; any other low-score residual is probed and extended only when the generated prefix clears the threshold (§4.2). The same τ_{pre} sits in front of ESC, Speculative Rejection, or DPTS (§4.2).

6 CONCLUSION

Disaggregated prefill moves every page the prefiller computed, including residuals a Tree-of-Thoughts fan-out will abort. Prefix caches retain those residuals until they expire. We introduced ForkServe, which prunes the fan-out before that transfer: a repeated loop is dropped before the kernel, any other low-score residual continues only after a short probe clears the threshold, and the connector inserts only the miss tail that remains. The copy-on-write tree is the substrate that makes the pruned residual the only new KV.

On Qwen3-8B GSM8K ($n=16$, $D=512$), $\tau_{pre}=0.15$ is the accuracy-preserving point in front of ESC, Speculative Rejection, and DPTS: 15/16, 16/16, 16/16 at $k=4, 8, 16$, a 23% cut in computed prefill (5008 to 3848 at $k=16$), and -5% , -6% , -17% end-to-end against native ESC (17.05 s to 14.14 s at $k=16$). $\tau_{pre}=0.45$ is the extra-speed point (-36% at $k=16$, 10.93 s, still 16/16) and loses two ques-

Table 15. Contest forest, $D=2048$, $k=4$, $tp=2$. MATH-500 is $n=200$; AIME is $n=73$. R1 is DeepSeek-R1; 4B is Qwen3-4B. Best storage and shortest end-to-end time in green.

Model	Method	peak / APC	acc.	e2e (s)
R1 MATH	APC	1863	—	0.560
	FS	1391	0.75	0.510
	FS+	1599	0.86	0.545
R1 AIME	APC	2118	—	0.055
	FS	1646	0.78	0.123
	FS+	1662	0.78	0.096
4B MATH	APC	1937	—	0.620
	FS	1770	0.91	0.625
	FS+	1770	0.91	0.600
4B AIME	APC	2150	—	0.027
	FS	1794	0.83	0.041
	FS+	1794	0.83	0.041

tions at small fan-out (14/16). On Qwen3-8B MATH-500 ($n=32$, $D=1024$), dropping every residual below τ cuts any-finished accuracy from 20/32 to 13/32 at $k=16$, while a 128-token probe restores 20/32 and lowers end-to-end time from 104.1 s to 72.2 s. On the Qwen3-14B GSM8K forest the spine is $0.72\times$ the median APC peak at matched accuracy (0.845 vs. 0.844). The earlier blanket-skip cost model cuts modeled fan-out prefill by 74.72% and raises the concurrency a fixed pool can sustain by 35.53%; that skip is not the GPU rule. Across six families the GSM8K spine stays at $0.72\text{--}0.75\times$ that peak, and the ratio falls from $0.89\text{--}0.90\times$ at $k=2$ to $0.53\text{--}0.55\times$ at $k=8$. At a 2048-token budget the ratio remains $0.75\text{--}0.91\times$, and ForkServe+ MATH-500 fan-out is 1284 ms against the prefix cache at 1593 ms on DeepSeek-R1 and 1014 ms against 1201 ms on Qwen3-4B. ForkServe’s key innovations include: (i) prefill pruning, which draft-skips repeated loops at $\tau_{pre}=0.15$ before any host decoder runs, then a probe that extends any other low-score residual only when the generated prefix clears the decode threshold; (ii) a disaggregated transfer gate that inserts only the miss tail of a residual that is kept, so aborted pages never cross the connector; (iii) a copy-on-write context tree that turns `fork` into an alias of the parent’s KV pages and frees only the residual on abort, so live occupancy is the spine $b(L + \ell_*)$; and (iv) longest-common-prefix commit together with a scheduler that admits speculative prefill only from the slack left by committed requests.

Future directions include probe lengths that track fan-out width, closed-loop QPS sweeps of the analytical concurrency gain, full-set Game24 and HumanEval forests, deeper trees and 32K-context windows, and multi-tenant admission with explicit latency margins on a shared frame pool. ForkServe shows that branch cost is incurred only for the residual that is committed, and only at the minimum-cost stage in the rejection cascade: the context tree and prefill admission together yield a spine whose occupancy is set by k and residual length, independent of tensor-parallel width. As agentic workloads scale, fan-out width, tree depth, and context length all grow against a fixed KV pool. Serving

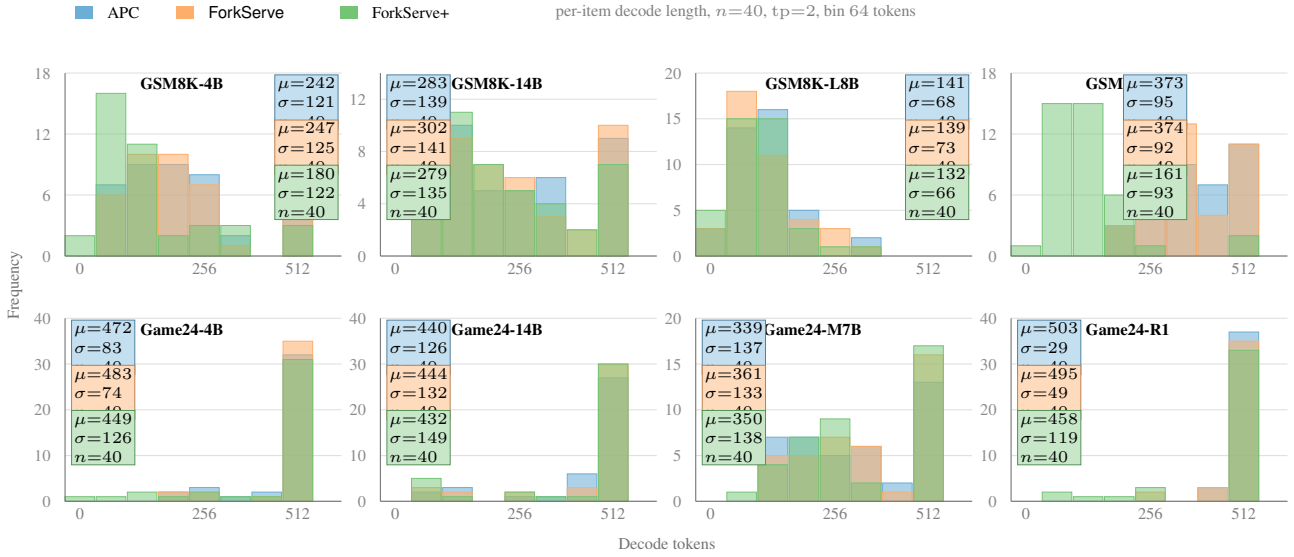


Figure 11. Per-item decode-length histograms ($n=40$, $tp=2$, $D=512$). Each panel overlays APC, ForkServe, and ForkServe+ (bin width 64). Boxes report μ , σ , and n . GSM8K (top) and Game24 (bottom) overlay APC, ForkServe, and ForkServe+. Llama-3-8B Game24 is omitted (all lengths 1); Mistral-7B is shown instead.

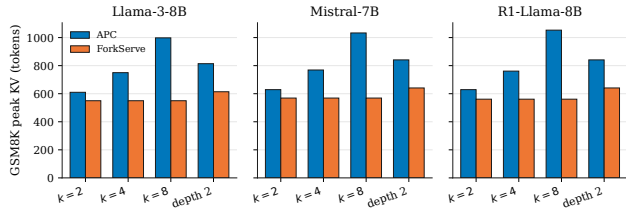


Figure 12. GSM8K peak KV versus branching ($n=16$, $tp=2$, $D=512$). ForkServe+ matches the ForkServe bar. The spine peak is invariant in k ; the APC peak grows with the hashed union, so the ratio falls from 0.89–0.90 to 0.53–0.55.

systems that fork by reference and prune before prefill retain capacity on the committed path because each new branch aliases the trunk, a residual is rejected at the first admissible stage, and abort returns only that residual, so the tokens that occupy HBM are the spine the decoder attends.

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